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Prof. Yang Shi
Editor-in-Chief
IEEE Transactions on Industrial Electronics

Dear Prof. Shi,

We are pleased to submit our original manuscript entitled “Primary-Side Control of Bipolar Field Excitation System for Pole-Changing Electrically Excited Synchronous Machines” for consideration as a *Letters to Editor* in *IEEE Transactions on Industrial Electronics*.

This work introduces a novel bipolar field excitation method for pole-changing EESMs. Unlike conventional WPT-based exciters that require receiver-side active converters and wireless communication, the proposed approach employs a dual-input, dual-output WPT system with anti-parallel rectifiers and a self-driven diode–MOSFET configuration. This enables bipolar excitation without any rotor-side active circuitry or wireless control signals. A small-scale prototype has been experimentally validated, achieving bipolar current reversal up to 2.88 A (118 W) with 99% rectifier efficiency. The concept simplifies the system and enhances reliability, features particularly valuable for traction drives.

The manuscript adheres to the IEEE format requirements, has not been published or submitted elsewhere, and has been approved by all authors. We appreciate your consideration and look forward to the reviewers’ feedback.

Sincerely yours,

Enes Ayaz
(on behalf of all authors)